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Integrating Environmental Impact into Cutting Speed Decisions in CNC Turning

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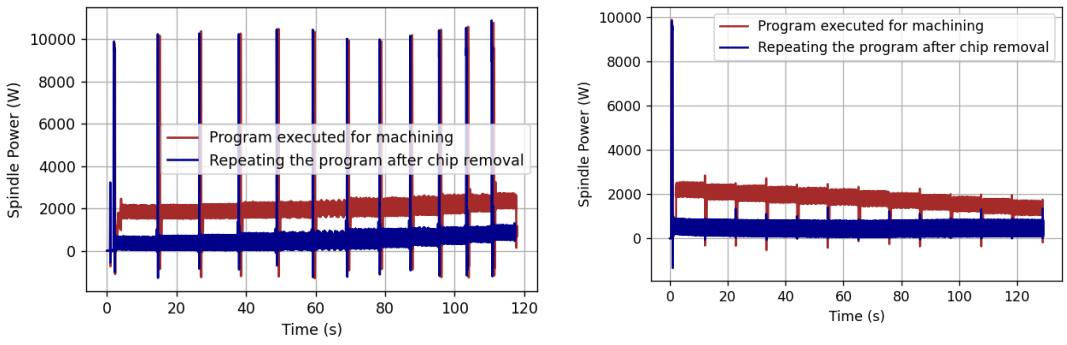
Abstract

In the manufacturing sector, there is a growing demand for environmentally sustainable practices that do not compromise production efficiency. Integrating smart sensors and providing access to CNC machine data enables the real-time monitoring of key process parameters. Machining operations consume energy not only for cutting, but also to maintain the spindle needed for cutting. This contributes to the carbon footprint (CO_2 emissions) of the manufacturing sector. This chapter presents the energy contribution to the Taylor analysis of the economic range, which is used to define the cutting speed of a turning operation on a CNC lathe.

Problem statement

The diameter needs to be reduced from 48 mm to 24 mm over a length of 40 mm on an AISI 4135 bar. The tool manufacturer recommended the following parameters for turning steel with DNMG NC5330 coated carbide tools: a_p = [0.4–4 mm], f = [0.1–0.4 mm/rev], and V_c = [90–200 m/min] for the depth of cut, feed rate, and cutting speed, respectively. The programmer used ap = 1 mm and f = 0.2 mm/rev. Having heard that changing the spindle speed on each pass consumes a lot of energy, he wrote two CNC programs on the machine tool: one using a constant cutting speed (V_c=150 m/min) and the other using a constant spindle speed (N = 1200 rpm) to check this. The Taylor coefficients for this pair of tool-workpiece materials are n = 0.24 and C = 230 m/min. These are used to determine the most economical option in terms of energy consumption, tool cost, and maximum production. The tool time change (t_{tc}) equals 1.5 minutes, and the machining cost is estimated at 1% of the tool cost ($\frac{C_m}{C_t} = 0.01$).

The objective of the problem is to recommend the best program to use, considering energy consumption, tool life, and production recommendations. The following must be calculated: the economic range using the traditional Taylor curve, the cutting power, cutting force, and specific cutting force, and the carbon footprint. Figure 1 shows the data collected using the CNC trace function during cutting and a second measurement without chip removal, and Table 1 summarizes the main data per pass[1].



(a) constant cutting speed (150m/min) (b) constant spindle speed (1200 rpm)

Figure 1 – Spindle power measured during both programs with and without chip removal

[1] The experimental data is available at: github.com/annacarlaaraujo/chapter_2025

Table 1: Average Spindle Power on each pass (W)

	V_c=150 m/min		N=1200 rpm	
ϕX, Diameter (mm)	Cutting	No Cutting	Cutting	No Cutting
46	1852.9	346.8	2211.8	470.5
44	1843.1	359.3	2112.7	452.9
42	1868.3	375.9	2036.7	442.2
40	1909.1	398.4	1961.5	437.5
38	1927.7	428.8	1892.4	429.6
36	1972.6	460.2	1827.3	427.2
34	2011.8	494.4	1755.3	425.0
32	2043.6	544.8	1681.2	422.5
30	2089.7	592.9	1608.1	421.4
28	2158.1	656.5	1534.0	421.4
26	2224.9	726.8	1462.3	420.1
24	2322.6	809.8	1388.6	419.5

Problem solution

Figure 2(a) shows the cutting geometry of the turning process, as well as the key parameters and specific situation described in the statement (Fig.2(b)).

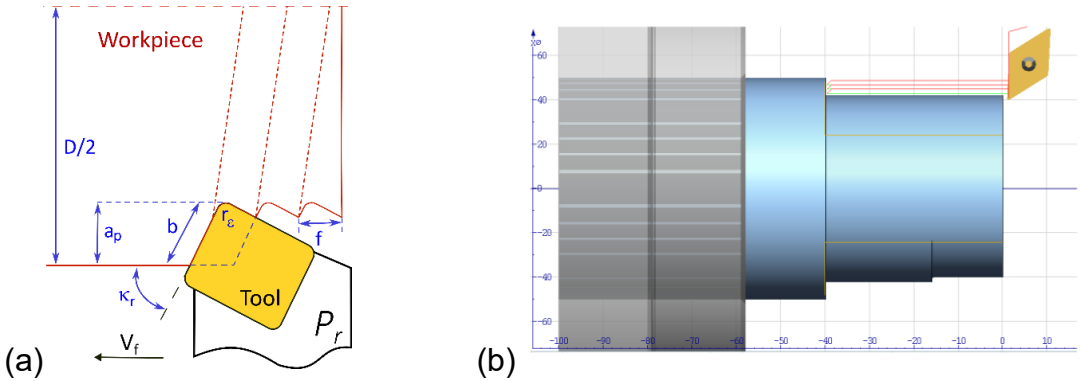


Figure 2. Turning process (a) Cutting geometry (a) Machine-tool simulation

To compare the velocities in both programs, the minimum and maximum spindle speeds in program I, with a constant cutting speed, are

$$\mathrm{N}[min, max] = \frac{1000\,\mathrm{Vc}}{\pi\, D} = \left[\frac{150000}{\pi.46}, \frac{150000}{\pi.24}\right] = [1037, 1989]\ \mathrm{rpm}\ (1)$$

In program (II), the minimum and maximum spindle speeds are:

$$V_c[min, max] = \frac{\pi DN}{1000} = \left[\frac{\pi.24.1200}{1000}, \frac{\pi.46.1200}{1000}\right] = [90.5,\ 173.4]\ \mathrm{m/min}\ (2)$$

The second step is to verify the machining economics interval (Gilbert, 1950), as parameters outside this range are not optimized. The traditional Taylor range for external turning is defined by the minimum-cost cutting speed and maximum production.

$$V_{cost} = \left(\frac{C.\frac{C_m}{C_t}}{(1-n)}\right)^{\frac{1}{n}} = \left(\frac{230\ 0.01}{(1-0.24)}\right)^{\frac{1}{0.24}} = 100.88\ \mathrm{m/min}\ (3)$$

$$V_{prod} = \left(\frac{C}{t_{tc}.n}\right)^{\frac{1}{n}} = \left(\frac{230}{1.5.0.24}\right)^{\frac{1}{0.24}} = 164.77\ \mathrm{m/min}\ (4)$$

It can be concluded that program, with V_c = 150 m/min, respects the economic range, whereas program II only respects the economic range for diameters between 28 and 42 mm. In other words, program II is not within the economic range at the beginning and end of the operation.

Now, based on the experimental spindle power presented in Table 1, let us calculate the cutting power (P_c), cutting force (F_c) and specific cutting force (K_c) based on the experimental spindle power presented in Table 1. First, it is needed to subtract the experimental spindle power without the chip removal ($\{SP\}_s$) must be subtracted from the measured spindle power during machining ($\{SP\}_m$) to calculate the cutting power P_c.

$$P_c(W) = SP_m - SP_s \ (5)$$

The cutting force F_c (N) and specific cutting force K_c (MPa) are calculated using Eq. (6) with the cutting parameters feed (f, mm/rev) and depth of cut (ap, mm). The results are presented in Table 2 for each program's pass.

$$P_c(W) = V_c.K_c.\ \mathrm{f}.a_p \ (6)$$

Table 2: Calculated cutting Power (W)

	Program 1 (Vc=150m/min)				Program 2 (N=1200 rpm)				
ϕX (mm)	Pc (W)	N	Fc(N)	Kc(MPa)	Pc (W)	Vc	Fc(N)	Kc(MPa)	MRR (cm^3/min)
46	1506.2	1038	602.5	3012.4	1741.3	173	603.9	3019.6	34
44	1483.8	1085	593.5	2967.6	1659.8	166	599.9	2999.6	33.2
42	1492.4	1256	597.0	2984.8	1594.5	158	605.5	3027.6	31.6
40	1510.7	1194	604.3	3021.4	**1463.2**	**150***	**609.4**	**3046.9**	**30***
38	1498.9	1256	599.6	2997.9	1462.8	143	613.8	3068.8	28.6
36	1512.4	1326	605.0	3024.8	1400.1	136	617.7	3088.4	27.2
34	1517.5	1404	607.0	3034.9	1330.3	128	623.6	3117.8	25.6
32	1498.8	1492	599.5	2997.6	1258.7	121	624.1	3120.7	24.2
30	1496.9	1592	598.7	2993.7	1186.7	113	630.1	3150.4	22.6
28	1501.6	1705	600.6	3003.1	1112.6	105	635.8	3178.8	21
26	1498.2	1836	599.3	2996.3	1042.2	98	638.1	3190.5	19.6
24	1512.8	1989	605.1	3025.6	969.1	90	646.1	3230.3	18

*MRR for 150 m/min=30 m.mm^2/min

The material removal rate (MRR) varies with the cutting speed, as calculated by Eq.7:

$$\mathrm{MRR}\ (cm^3/min) = V_c.\ \mathrm{f}.a_p \ (7)$$

The energy consumption E_{SP} (J) per pass, considering the total spindle power SP_m is:

$$E_{SP}(J) = SP_m(W).t_c(s) = \frac{SP_m.L_c.60}{f.N.1000} \ (8)$$

Specific energy consumption (SEC) is the amount of energy required for a machine tool to remove 1 cm^3 of material (Eq. 9), and it is a well-known key performance indicator (KPI) for environmental analysis. Once the calculation has been performed, it is possible to analyze the evolution of power and energy consumption per pass. Table 2 can be completed, but we have chosen to present the results graphically in Fig. 3.

$$\mathrm{SEC}(J/cm^3) = \frac{E_{SP}}{Vol} = \frac{SP_m.t_c}{MMR.t_c} = \frac{SP_m}{MMR} \ (9)$$

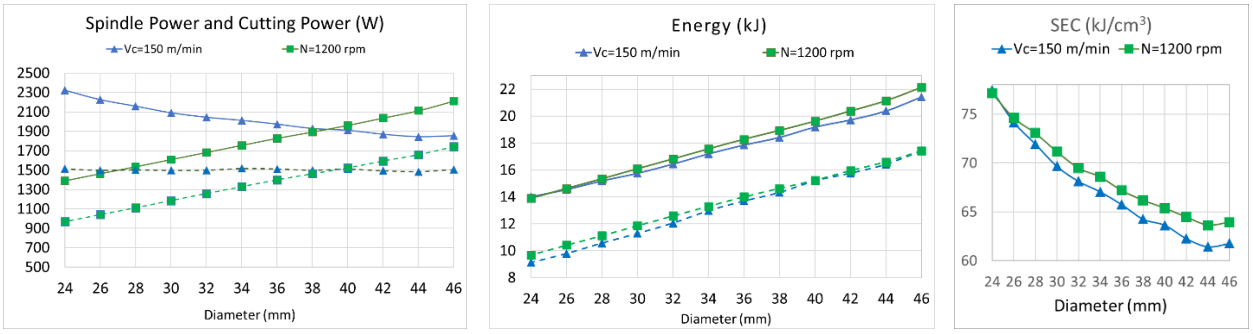


Figure 3 – Power, energy, and specific energy consumption.

As can be seen, the spindle power consumed does depend on the diameter in programs I and II. In program I, the difference is related to the power consumed for turning the piece rather than for machining. However, when multiplied by the cutting time, which depends on the feed velocity, the energy consumed is the same in both programs. The difference in energy consumption between cutting and air cutting remains constant and is independent of the cutting parameters. Program II is also more interesting in terms of energy consumption, as all SPE values are lower than in Program I.

The carbon emissions resulting from generating this amount of electricity for a CNC machine tool are estimated based on the grid. Taking the South China grid (Li et al., 2015) as a reference, EF = 0.6 kg CO_2/kWh. It is therefore possible to calculate the carbon footprint in terms of CO_2 emissions (Em_{CO2}) in (kg) consumed by Program I per piece:

$$\mathrm{Em}_{CO2} = \sum E_{SP}(kJ)/3600\ (s/h).\ \mathrm{EF}(kg\mathrm{CO}_2/kWh) = 12\ * 17.9 * 0.6/3600 = 0.04 kg\mathrm{CO}_2$$

Conclusions

In this chapter, carbon emission is calculated using experimental data for both cases: constant spindle speed and constant cutting velocity. Although adapting the spindle speed raises average power consumption, maintaining the cutting speed remains the best option. Additionally, it is well known that changing the cutting velocity affects tool wear. Still, reducing non-cutting energy use is important, as it does not add value to the workpiece.

Reference

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